Overview of Economic Valuation Methods

Table 1 presents a number of widely-used economic valuation methods. It indicates which aspects of nature they are suited to value, if they measure preferences, and whether they can capture both use and non-use values.

Table 1 Economic valuation methods

Method	Suitable for	Value type	Consumer surplus	Cost/benefit-based	Use versus non-use value
CVM	Amenities and other perceived attributes of natural sites	Willingness to pay	Included	Benefit-based; Stated preference	Both
HPM	Amenities/nuisances related to capital goods	Hedonic price	Included	Benefit-based; Revealed preference	Use
TCM	Recreational value of natural sites	Travel costs	Included	Benefit-based; Revealed preference	Use
PFM	Natural qualities that influence economic production	Market price Economic rent	Excluded	Cost-based; damage costs Revealed preference	Use
ABM	Natural qualities effecting consumer behaviour	Prevention costs	Excluded	Cost based; prevention costs Revealed preference	Use
Acronyms: CVM = Contingent Valuation Method; HPM = Hedonic Pricing Method; TCM = Travel Cost Method; PFM = Production Factor Method; ABM = Averting Behaviour Method.					

This table only includes valuation methods that measure preferences. There are also several economic valuation methods that are not based on social preferences, such as Energy Analysis and Implicit Valuation (Hoevenagel, 1994). These non-preference based methods do not really measure the welfare provided by nature and are therefore not elaborated upon here.

The Contingent Valuation Method (CVM)

CVM is a survey method in which respondents are asked how much they are willing to pay for the use or conservation of natural goods. Their stated preferences are assumed to be contingent upon the alternative goods that are offered in a 'hypothetical market'. Essential elements of the survey are: description of the natural good that is to be valued, description of the payment vehicle and description of the hypothetical market. *Describing the natural good* includes identifying all valuable attributes of the good. The *payment vehicle* pertains to how the money will be paid. One can pay for a good in cash every time it is used or by means of an increased income tax. The *description of the hypothetical market* should include an identification of who will provide and who will pay for the nature improvement. It should be made clear that the payment is a collective action; everybody else will also pay, otherwise respondents may refuse to pay although they appreciate the good. Respondents should also be reminded of the possibility of spending their income on goods other than nature, to prevent overestimates (Hoevenagel, 1994).
CVM measures benefit-based preferences and it includes the consumers' surplus. It is said to be an appropriate economic valuation method for environmental goods that have no indirect effects on other goods. It is therefore suited for the valuation of amenities or other easy to perceive aspects of nature, but not for the valuation of natural processes, such as climate regulation, where effects on human welfare are difficult to grasp. However, CVM does not produce valid measurements when it concerns goods that people are not familiar with. Nor does it work when people reject responsibility for the good in question (de Boer *et al*., 1997). If people are asked, for example, about their willingness to pay for clean soil, they may state that it is zero, because they feel the polluter should pay. This does not mean that they do not appreciate clean soil. One may also remark here that it is better to value goods that have an international character on a cost basis, because in a CVM-survey respondents will not know what to answer if they realise that reducing pollution in their own country does not solve the global pollution problem, if the other countries do not make an effort too.
Several authors (Freeman, 1986; Hoevenagel, 1994; Wierstra *et al.*, 1995) argue that CVM works best where it is least needed. It does not provide valid estimates when people are unfamiliar and inexperienced with the good. Validity may be a problem, since it is very difficult to describe a natural good in such a way that all its attributes are accounted for. Freeman (1986) notes that CVM works best for those goods resembling ordinary commodities. This means that it is best suited for valuing consumption goods that people consume more of as their income increases (and the price decreases). When goods are not easily commoditised, as in choices concerning entire ecosystems, CVM results are doubtful. Supporters of this notion argue that an environmental good has several attributes and that compressing the values of these attributes into a single metric (such as willingness to pay) leads to an information loss. The same argument is, however, also relevant to private goods, but in that case people seem to accept their prices as their true value. The idea behind this is that people have experience in valuing and making trade-offs for attributes of private goods, whereas they do not have any experience in valuing environmental goods. In fact, they may not even be aware of all attributes. This situation makes people liable to construct their values heuristically on the basis of the information provided by the elicitation setting (Vatn and Bromley, 1994). Taking account of the restrictions of valuation experience and commodity resemblance, it seems that CVM is most accurate for the valuation of publicly-managed goods with private characteristics such as natural sites.
In CVM surveys one can encounter various sources of bias, such as samples which are not representative, strategic behaviour of respondents or confusion about the size of the good that is to be valued (part-whole bias).
Although CVM studies are known to have large data requirements, they do not require secondary data. Since one can gather all necessary data, i.e. the willingness to pay and its explaining variables such as income and attitude towards nature, by means of one survey (Hoevenagel, 1994), CVM seems suited for gathering input data for a decision support tool.
Since CVM is only suited for 'easy to perceive aspects' of the natural environment, one may conclude that the method does not allow us to determine the total economic value of nature. However, CVM can be used to estimate the non-use value of nature.

The Hedonic Pricing Method (HPM)

HPM is based on the idea that market goods are often traded at prices in which amenities are internalised. For example, the price of a house in quiet and beautiful surroundings is likely to be higher than the price of the same kind of house next to a smelly factory. The HPM starts with a regression of house prices (or wages) against all their valuable characteristics. This leads to a hedonic price function of the following shape: Value(house) = F(architecture, contents, amenities, local taxes, noxious facilities, etc.). From this function one can calculate the willingness to pay for a marginal change in each of these explaining variables. This is the implicit price of the amenity under investigation. From these implicit prices, the demand curve for a specific amenity can be derived. The demand curve is then used for estimating the economic value of an amenity such as natural beauty.
The HPM has often been used to measure the (negative) values of noxious facilities and for the value of environmental goods such as air quality improvement. It has also been applied to value job risks and to value natural sites or parks that might have a positive influence on house prices. The HPM measures revealed preferences and it includes the consumers' surplus as it measures the total area underneath the derived demand curve. There are, however, several arguments which shed doubt on the validity of the results of the method.
The validity of the method may be questioned because the shape of the hedonic price function is not known. It is also possible that there are several amenities that influence the price of a house in opposite directions. There may, for example, be a positive influence of a park nearby, but at the same time two noxious facilities which supply jobs. It is also possible that the house market is distorted due to governmental interventions (Pearce and Markandya, 1989). Since the number of explaining variables can be numerous, one runs the risk of not including an important variable or encountering multi-collinearity and thus drawing spurious conclusions about the value of an amenity.
HPM has a very large data requirement because both primary data (characteristics of the surroundings) and secondary data (market transactions) need to be collected The value of a house or wage depends on many factors: there are social factors, such as employment opportunities, taxes and accessibility. Data need to be gathered for all these factors. This makes HPM less suited for application in a decision support tool.
A possible source of bias is the representativeness of the sample for the population. The value of, for instance, parks may be underestimated, because people who do not live near a park may also appreciate it. Their appreciation is, however, not reflected in the price of their house and they are not included in the sample (de Boer *et al*., 1997).
Although HPM can be used to value amenities such as natural beauty, this may not be enough to capture the total economic value of a natural site. Beauty is only one attribute of a natural site. All other welfare-generating characteristics of the site, such as biomass production cannot be captured by this method. The HPM was not developed to determine the total value of nature, but to determine the value of amenities only. HPM cannot be used to determine non-use values of nature.

The Travel Cost Method (TCM)

TCM relies on the assumption that people make repeated trips to recreational sites until the marginal utility derived from a trip equals the marginal costs of a trip. The marginal costs are travel costs in terms of time cost and transportation cost. These travel costs can be regarded as a directly revealed preference for recreation and an indirectly revealed preference for nature.
The TCM assumes that the demand for trips to a specific site is dependent on travel costs, income, characteristics of the site, prices of substitutes, etc. Travel costs are, however, related to distance. In order to determine the willingness to pay of visitors from various distances, distance circles are drawn in the service area of a site. The percentage of inhabitants of each circle that will visit the site at a fixed amount of travel cost per circle is determined by means of a survey. Now it is assumed that if, say, 25 % of the population in the second circle will visit the site at a cost price of, say, $10, while 50 % of the population of the first (closer) circle will visit at a cost price of $5, then the actual willingness to pay of a quarter of the visiting population in the first circle is $10, even though they only have to pay $5 now because they live nearby. In this way a

demand curve for the site is derived. The TCM assumes that people in all circles have homogeneous preferences.
TCM measures revealed preferences for natural sites in terms of willingness to pay for site visits. As such, it includes the consumers' surplus. The TCM has often been applied on water recreation, wetland visitations and hunting.
The validity of TCM can be doubted because it is unclear whether the method actually measures preference for nature. People might not have been aware of the total pleasure or costs before they decided to go on a trip. They may also combine a visit to a natural site with a family visit (multipurpose trip). It is debatable whether the full travel cost to a natural site may be attributed to nature, because these expenditures may not have been made purely for nature. One could argue that the expenditures were also made for the enjoyment of the transportation, leaving less money to spend on nature. In fact, attributing the full costs of services, such as travel, to the final products they were aimed at such as nature, means that the value of these services is reduced to zero. This does not seem compatible with reality.
A bias of TCM concerns the representativeness of the sample. People who do not visit a site may still value the site, but they are excluded from the sample. On the other hand, people who really appreciate the site may have very low travel cost, because they have moved to live closer to the site. Population distribution (far from/close to the site) has an impact on the demand for recreation and thus on the valuation of the site. The strength of this impact depends on whether one is looking at the total value of a site or at the value per visit. Most technical problems with TCM relate to this (Stynes, 1990).
The data requirements of TCM are rather large, since one has to interview visitors of sites to determine their travel costs. One also has to determine the percentages of subpopulations within the service area of a site that will visit the site.
TCM can only capture part of the total economic value of nature. The recreational value is only part of the total value of nature, since society does not only derive welfare from nature through recreation. A consequence of the fact that TCM only reveals the welfare realised by the visiting population is that it can only measure use values.

The Production Factor Method (PFM)

PFM is based on the fact that many natural resources, processes and qualities are used as production factors. Improvement of natural quality may lead to a reduction of production costs for the sector making use of the relevant quality. The PFM tries to value natural qualities by valuing their impacts on production costs. The method consists of two steps: first, the relation between a dose of pollution/pressure and an effect on production (the response) is determined. Secondly, the response must be translated into economic terms. This is can be done by means of the previously described valuation methods (de Boer *et al*., 1997) and consequently the PFM is not really a separate valuation method. The valuation part of PFM is mostly done by simply multiplying the quantity change by the market price. It would, however, be better to investigate all economic effects, such as changes in demand and supply and in prices.
The PFM has been applied to value the effects of water quality on agriculture, fishery and industry and the effects of air quality on buildings, crops and livestock. The method is meant to determine the value of changes in natural qualities on the economic production system. Consequently, it is not directly suited to determine the economic value of natural sites which are not cultivated.
Due to the use of market prices to value certain environmental qualities, the validity of the PFM is doubtful. PFM can only produce cost-based estimates of the value of the production capacity of nature and it does not include consumers' surplus when market prices are used. The costs of production losses may not cover the full social preference for nature.
A bias of the PFM is that effects on production may have been distorted by averting behaviour. Producers will try to avert the effects of reduced natural qualities by undertaking all sorts of prevention activities, such as shifting to different crops or products, adapting cultivation or harvesting techniques. They may even have left the market to become involved in other types of economic production.

Because PFM is based on dose-response relations which involve a considerable amount of ecological information and because it requires economic data on natural products as well, it has a large data requirement. If one wishes to account for demand and supply dynamics in the valuation of responses, data requirements will only become larger.
PFM allows one to determine the value of the production capacity of nature. For example, a reduction of water quality may cause the fish population to decline and consequently reduce the income from fisheries. At the same time the poorer water quality may cause a rise in the production costs of drinking water (i.e. increased treatment costs). The total economic value of nature does, however, comprise more attributes than fish production and water purification, and therefore the PFM can only capture part of the total economic value of nature. The PFM cannot capture the non-use value of nature.

The Averting Behaviour Method (ABM)

ABM is especially suited for valuing natural qualities. This is done by looking at expenditures made to avert or mitigate negative effects from the reduction of a natural quality. ABM relies on the assumption that people perceive the negative effects of environmental deterioration on their welfare and that they are able to adapt their behaviour to avert or reduce these effects. This means, for instance, that people experience the negative effects of ozone depletion and that they will buy products such as hats and suncream to prevent damage to their health. The willingness to pay for a clean environment is deducted from people's purchases of products and services to avert the negative effects of pollution.
Most applications of the ABM concern the purchase of protective items in relation to an environmental quality. As such, ABM is a cost-based method, since the costs of purchasing these items are used to value environmental qualities, even though the social preferences for a healthy environment may be much greater than the expenditures on these products. Since the market prices of products are used to value the environment, this method cannot capture the consumers' surplus.
The validity of ABM is questionable because of the assumption that people actually purchase certain goods to protect themselves against environmental deterioration. In fact, people may make the purchase for the sake of the product (in the case of suncream it would be cheaper to stay away from the sun) and not to avert negative effects from their living environment. Buying such products actually means that they have less money to spend on the environment.
Another concern with ABM is that people may not react to small changes in environmental quality. They may only react when a certain threshold has been passed. Besides that, one may remark that such defensive purchases will not wipe out and will certainly not reverse the negative effects of environmental quality deterioration.
Determining the value of a clean environment in terms of the expenses made to avert negative effects of environmental deterioration, automatically introduces a bias; only the willingness to pay of people who display certain averting behaviour is measured, while people who behave differently may also have a willingness to pay for a clean environment.
The data requirements of ABM may be large, depending on the number of ways in which people try to avert the effects of an unhealthy environment. On top of that, health-behaviour and health-environmental quality relations are difficult to determine and are not easy to verify in practice.
Since ABM is mainly oriented toward health risks, it can only capture the value of certain attributes of nature (clean air in this case) but it cannot be used to determine the total economic value of nature. It cannot be used to determine non-use values.

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